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Mixing of menthol-based hydrophobic deep eutectic solvents as a novel method to tune their properties

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ABSTRACT

Due to the limited number and water-immiscibility of hydrophobic deep eutectic solvents (hDESs), new methods of expanding the repertoire and tuning the properties of these solvents are required. Here, menthol was combined with organic acids or alcohols to produce nine hDESs, all of which were freely miscible with each other. Three of the hDESs, vis. ML, MB, and MT (containing menthol combined with levulinic acid, 1,3-butanediol, and *tert*-butanol, respectively) were selected as the least to most non-polar solvents. Two and three of these hDESs were mixed at equal volume ratios to produce three binary mixtures (namely, LB, BT, and LT) and one ternary mixture (LBT), respectively, via simplex centroid mixture design. The properties of these mixtures as extraction solvents were characterized using *Ginko biloba* leaves (GBLs) as a model plant. Mixtures containing ML exhibited high capability for extracting phenolic compounds and flavonoids, comparable to that of methanol and 70% methanol. They provided wide extraction coverage with reliability, as evidenced by the high numbers of variables with coefficients of variation within 30%. The liquid chromatography-mass spectrometry-based profiles of 47 secondary metabolites showed that MB and BT presented similar results to methanol and 70% methanol based on heatmap and hierarchical cluster analysis. This study shows that combining the menthol-based hDESs furnishes properties that differ from those of the individual components; in this way, the repertoire of hDESs can be expanded, and therefore, the hDESs and their mixtures can be used as green task-specific extraction solvents as substitutes for volatile organic solvents.

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1. Introduction

Solvents are extensively studied in green chemistry due to their extensive usage in diverse applications in chemical processes [1]. Although water is the greenest solvent, its negligible solubility for most organic compounds limits its usage [2]. Thus, volatile organic solvents that are often toxic, environmentally hazardous, and have low biodegradability are still used dominantly in most applications [3]. The development of new and safe green solvents has been in continuous demand from the perspective of the impact on human health and the environment [4].

Deep eutectic solvents (DESs) are recognized as green solvents due to their low volatility, non-flammability, high biodegradability, and designability [5]. DESs are generally prepared from naturally occurring components, including a large number of hydrophilic compounds, such as sugars, organic acids, sugar alcohols, and amino acids. In general, DESs are freely miscible with water, and their properties can be modulated by adding water [6]. A variety of DESs show high solubility for a wide range of compounds [7] and have been successfully used to extract various classes of natural products, including phenolic compounds [8], flavonoids [9–12], ginsenosides [13], and terpenoids [14]. DESs also have been used in diverse applications such as in the preparation of stationary and mobile phases for chromatography [15] and removal of pollutants [16,17].

Recently, DESs that are hydrophobic and generally immiscible with water, so called hydrophobic DESs (hDESs), have been introduced as an alternative to water-immiscible hazardous organic solvents [18]. hDESs have been used to extract organic and inorganic analytes, including volatile aromatic hydrocarbons [19], pharmaceuticals [20], heavy metals [21], and pigments [22], from various types of matrices [23]. Quaternary ammonium salts and volatile monoterpenes have been used as the major components of hDESs. Menthol has been preferentially used to synthesize diverse hDESs as it allows facile preparation, and the resulting menthol-based hDESs have lower viscosity than quaternary-ammonium-based hDESs [24]. Compared with hydrophilic DESs, however, the number of hDESs reported thus far is limited due to the restricted repertoire of hydrophobic components available for hDES synthesis. Moreover, the water-immiscibility of hDESs (although a small portion of water may dissolve in hDESs [25]) means that their properties cannot be tuned using water. Instead, a ternary hDES system composed of three components was developed [26,27], and the types

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and molar ratios of HBA and HBD were varied to tune the properties of hDESs [28]. Nonetheless, the repertoire of hDESs must be expanded, and novel methods of tuning their properties are required to make hDESs more accessible for diverse applications.

In this study, it is hypothesized that the solvent properties of hDESs can be modulated by mixing different hDESs, as in the case of conventional solvents having different properties. To test this hypothesis, a series of menthol-based hDESs is prepared, and three kinds of hDESs with relatively different polarities are applied to a simplex centroid mixture design. The resulting menthol-based hDESs and their mixtures are carefully characterized as extraction solvents in multifarious aspects. For this purpose, *Ginkgo biloba* (GB) is selected as a model system for extraction. There are extensive studies on GB, including evaluation of the bioactive compounds and their bioactivities [29,30]. DESs were used to extract aromatic acids [31], flavonoids [32], proanthocyanidin [33], terpene trilactones [34], and polyprenyl acetates [26] from GB. More relevantly, several studies reported the use of hDESs with different polarities to extract various phytochemicals in GB. For instance, a two-phase DES system consisting of hydrophilic and hydrophobic DESs was used to simultaneously extract various bioactive compounds with different polarities [35], and a two-phase aqueous micellar system based on menthol-based hDESs was used to separately extract many bioactives [36]. Here, the efficiency of the prepared solvents for the extraction of a few classes of bioactive compounds is evaluated. The extraction coverage with reliability and the chemical profiles of the secondary metabolites are also analyzed. For the first time, this study demonstrates that the properties of hDESs can be tuned by mixing several hDESs and that this strategy can be exploited to expand the repertoire of hDESs as substitutes for toxic volatile organic extraction solvents.

2. Materials and methods

2.1. Chemicals, reagents, hDES synthesis, and sample preparation

The chemicals and reagents used in this study are listed in Table S1. The other chemicals not listed in Table S1 were acquired from Sigma-Aldrich (St. Louis, MO, USA) unless elsewhere stated. Menthol and its counterparts were directly used without purification to prepare a series of hDES by mixing in a molar ratio of 1:2 and heating as previously described [37]. The resulting hDESs are listed in Table 1.

Ginkgo biloba leaf (GBL) samples were provided by Amorepacific Corporation (Seoul, Korea) [11]. The leaves were pulverized by an electric mill and stored in a sealed container at 4 °C until use. The finely powdered samples were directly used for extraction without sieving. Fifty milligrams of powder was ultrasonicated in 1 mL of the extraction solvent for 60 min. After centrifugation at 20,238*g* for 10 min, the supernatant was transferred and diluted in methanol for subsequent use. The solvent information is presented in Table 1. The control extraction solvents include four conventional solvents (water, methanol, 70% aqueous methanol, and ethyl acetate) and three hDES components (levulinic acid, 1,3-butanediol, and *tert*-butanol).

2.2. Measurement of the solvent polarity, water solubility, and capacity for extraction of bioactive compounds

The relative polarities of the menthol-based hDESs were estimated using Nile red as a solvatochromic probe [38]. In brief, 200 μL of 10 μg mL^{-1} Nile red in ethanol was added to 1 mL DES or a referent solvent and vortexed vigorously. The absorption spectra and λ_{max} values were analyzed to determine the molar transition energy (E_{NR}), as described previously [39]. The Kamlet-Taft solvatochromic parameters were estimated using 4-nitroanisole, 4-nitroaniline, and 4-carbomethoxy-1-ethylpyridinium iodide as the probes for polarizability (π^*), H-bond acceptor ability (β), and H-bond donor ability (α), respectively, according to the literature [40]. The water content of each DES after shaking with water was measured using a Metrohm Karl Fischer coulometer.

Table 1
Polarity of the menthol-based hDESs and their mixtures in comparison to reference solvents.

Solvent			Molar ratio	E_{NR} (kcal mol^{-1})	Abbreviation
hDESs	Component 1	Component 2			
	Menthol	Levulinic acid	1:2	50.34	ML[a]
		Propionic acid	1:2	51.70	MP
		1,3-Butanediol	1:2	51.74	MB[a,c]
		1,2-Octanediol	1:2	51.98	MO
		Dipropylene glycol	1:2	52.04	MDG
		Isopropanol	1:2	52.20	MI
		1-Undecanol	1:2	52.30	MU
		1-Dodecanol	1:2	52.43	MD
		tert-Butanol	1:2	52.65	MT[a,c]
hDES mixtures (1:1, v/v)	ML + MB			51.36	LB
	ML + MT			51.83	LT
	MB + MT			51.95	BT[c]
	ML + MB + MT			51.83	LBT
Reference solvents	70% Methanol[c]			50.52	
	Methanol[c]			51.70	
	Ethyl acetate[c]			54.32	
	Levulinic acid			N/A[b]	
	1,3-Butanediol[c]			50.96	
	tert-Butanol[c]			52.88	

[a] Selected for the solvent mixing.
[b] Not available due to difficulty of correct measurement caused by its acidity.
[c] See Table S2 for the Kamlet-Taft parameters.

The total phenolic content (TPC) of the extracted phase was determined as described by Jeong et al. [41] with slight modifications. Briefly, 200 μL of a 50-fold diluted extract was mixed with an equal volume of 0.2 N Folin–Ciocalteu (F–C) reagent and incubated in the dark for 5 min. The reaction was quenched by adding 800 μL of 20% Na_2CO_3 and incubated for 15 min without light. The absorbance of the supernatant was determined at 725 nm. Gallic acid showed a linear response in the range of 2–50 μg mL^{-1} ($y = 0.02083\times - 0.00686$, $r^2 = 0.9995$). The TPC is expressed as gallic acid equivalents (mg GAE/g GBL).

The method presented by Chupin et al. was used to measure the total flavonoid content (TFC) [42]. Equal volumes of 50-fold diluted extract and 2% $AlCl_3$ in methanol were mixed and incubated at room temperature for 10 min. The absorbance was measured at 415 nm. Quercetin, used as a standard, showed a linear response in the 2–50 μg mL^{-1} range ($y = 0.02436\times - 0.03553$, $r^2 = 0.9996$). The TFC is expressed as quercetin equivalents (mg QE/g GBL).

2.3. Non-targeted metabolic profiling of GBL extracts

Non-targeted metabolic profiling of the samples was conducted using an ultra-high performance liquid chromatography-quadrupole-time-of-flight mass spectrometry (UHPLC-Q-TOF/MS) system [43]. An Agilent ZORBAX Eclipse Plus C_{18} column (100 mm × 2.1 mm, 1.8 μm) was eluted with the mobile phase consisting of 0.1% formic acid in water (A) and 0.1% formic acid in acetonitrile (B). Gradient elution was performed as follows: 0–5 min, 5–50% B; 5–20 min, 50–100% B; 20–24 min, 100% B; 24–24.5 min, 100–5% B. The injection volume, flow rate, and column temperature were set as 5 μL, 0.2 mL min^{-1}, and 30 °C, respectively. The MS system was operated in both negative (NEG) and positive (POS) ionization modes with a data acquisition range of 50–1200 *m/z*, while the other system parameters were the same as used in a previous study [43]. The clear supernatant (see Section 2.1) was diluted 10-fold and filtered through a 0.2-μm membrane filter (Millipore, Tullagreen, Ireland). Quality control samples were prepared by mixing equal volumes of all samples and were

analyzed every eight samples in the running sequence to monitor the system stability.

The metabolites were putatively identified by matching the detected mass to the theoretical mass from PubChem (http://www.ncbi.nlm.nih.gob/), Human Metabolome database (http://www.hmdb.ca/), and the literature. The identities were further confirmed using the corresponding available standards.

2.4. Data processing, experimental design, and statistical analysis

The UHPLC-Q-TOF/MS data were processed using Waters Markerlynx XS 4.1 software (Waters, Milford, MA, USA). The data processing parameters were set as follows: XIC window, 0.02 Da; marker intensity threshold, 500; mass window, 0.05; and retention time window, 0.20. The resulting datasets included all detected variables from each sample with the mass, retention time, and corresponding normalized intensities. The normalized intensity of the peaks of the identified compounds was converted to the z-score and processed in Multiexperiment Viewer (MeV 4.9.0; http://tm4.org/mev/) for heatmap and hierarchical cluster analysis (HCA). The Manhattan distance metric and average linkage with optimization of the sample leaf order were used in HCA.

The extraction coverage of the solvent was assessed in terms of the number of variables (NV) with a coefficient of variance (CV) ≤ 30%. The common variables between different solvents were expressed in a Venn diagram using an online tool (http://bioinformatics.psb.ugent.be/webtools/Venn/).

Design Expert 8.0 software from Statease Inc. (Minneapolis, MN, USA) was used for simplex centroid mixture design, which allowed the generation of seven different combinations of three menthol-based hDESs and evaluation of their TPC and TFC extraction ability. Multiple comparison based on one-way analysis of variance (ANOVA) followed by Tukey's post hoc analysis was conducted using GraphPad Prism 6 for Windows (GraphPad Software, San Diego, CA, USA).

3. Results and discussion

3.1. Preparation of mixtures of menthol-based hDESs with varying polarity

Nine organic acids and alcohols that could form eutectic mixtures with menthol based on previous reports [7,18] were selected as the hDES components. These compounds include levulinic acid, propionic acid, 1,3-butanediol, 1,2-octanediol, dipropylene glycol, isopropanol, 1-undecanol, 1-dodecanol, and *tert*-butanol, which resulted in the following nine menthol-based hDESs, ML, MP, MB, MO, MDG, MI, MU, MD, and MT, respectively (Table 1). Consistent with the literature [44], all the prepared DESs showed sufficiently good fluidity for handling. All the hDESs were fully miscible with each other.

The transition energy (E_{NR}) of the nine hDESs, determined using Nile Red as the solvatochromic probe, was relatively high (50.34–-52.65 kcal mol^{-1}; Table 1), indicating that the solvents are much less polar than water (48.24 kcal mol^{-1}). The polarity differed depending on the DES components. Herein, three hDESs, ML, MB, and MT, were selected as the least, medium, and most non-polar solvents, respectively, to produce solvent mixtures for the ensuing analyses. The polarity of the three counterparts of these hDESs, levulinic acid, 1,3-butanediol, and *tert*-butanol, all of which were liquid at room temperature, followed the order: levulinic acid > 1,3-butanediol > *tert*-butanol. The polarity of the corresponding hDESs followed the same order. Moreover, the relatively high polarity of levulinic acid and 1,3-butanediol, and the relatively non-polar nature of *tert*-butanol were mitigated by menthol upon formation of the hDESs as indicated by the E_{NR} values.

The three hDESs were mixed in equal volumes based on a simplex-centroid mixture design, resulting in three mixtures of two hDESs, namely LB, LT, and BT, and one mixture of three hDESs, LBT. As shown in Fig. S1, all of these solvents could be phase-separated from water with slight to moderate dissolution to water; they generally remained stable for several days. The phase-separated solvents contained 1.9–6.4 wt% water. These suggest that these solvents could be used in the aqueous extraction environment. The E_{NR} values of these mixtures showed that the polarity of the hDES mixtures was intermediate between that the component hDESs (Table 1). For example, the E_{NR} value of LB, comprising ML ad MB, was 51.36 kcal mol^{-1}, which is intermediate between those of ML (50.34 kcal mol^{-1}) and MB (51.74 kcal mol^{-1}). Several reference solvents were also tested for comparison. Aqueous 70% methanol [30] and methanol [45] are not only frequently used as extraction solvents, but also their polarity (50.52 kcal mol^{-1} for 70% methanol and 51.70 kcal mol^{-1} for methanol) is close to that of ML and MB (Table 1). Besides these water-miscible solvents, ethyl acetate, which is water-immiscible and often used in plant extraction [46], was also tested, although its polarity (54.32 kcal mol^{-1}) is lower than that of any of the other hDESs tested in this study. The Kamlet-Taft parameters of the selected hDESs and their mixtures were further measured and summarized in Table S2; the solvents containing levulinic acid could not be characterized because the probes were incompatible with acids. The acidity (α) of the solvent mixture, BT was between that of MB and MT, as similarly found in E_{NR}. However, the polarizability (π^*) of BT was lowered and the basicity (β) was elevated compared to those of MB or MT. These results imply that the hDESs mixtures could have solvent properties different from the component hDESs.

3.2. Efficiency of menthol-based hDES and their mixtures for extraction of bioactive compounds

The efficiency of the individual hDESs and their mixtures for the extraction of two representative classes of bioactive compounds (TPC and TFC) from GBL (Table S3 and Fig. 1) was evaluated. Two significant models with insignificant lack of fit and high coefficients of determination could be established for TPC and TFC; the corresponding ANOVA data and model equations are provided in Tables S4 and S5, respectively.

For extraction of the total phenolic compounds, 70% methanol, methanol, and levulinic acid, which are relatively polar, were the most effective among the reference solvents. In contrast, the two most non-polar solvents, ethyl acetate and *tert*-butanol, were the least effective and were even inferior to water (Table S3 and Fig. 1a). The TPC extraction efficiency of the hDESs with 1,3-butanediol and *tert*-butanol as counterparts to menthol (MB and MT) were significantly higher than that of the individual solvents without menthol. In fact, all of the hDESs and their mixtures presented similarly high TPC levels. The TPC levels of ML and LB were particularly high compared to that of the other hDESs. The response surface contour plot shows that the highest TPC extraction could be achieved with ML and its mixtures (Fig. S2a).

The TFC extraction efficiency of the reference solvents and the hDESs varied greatly, with methanol and ML showing the highest values (Table S3 and Fig. 1b); all of the solvents were superior to water. Similar to the TPC results, the TFC extraction efficiency of ML, MB, and MT were significantly higher than that of levulinic acid, 1,3-butanediol, and *tert*-butanol, respectively ($p < 0.01$). Levulinic acid appeared to be an effective DES component for the extraction of phenolics and flavonoids, given that ML and its mixtures had reasonably high TPC and TFC extraction levels. This is plausibly attributed to the acidity of levulinic acid, where acidic conditions have been shown to be favorable for flavonoid extraction [47]. The statistical model suggests that the 2:1 mixture of ML and MB should yield the highest TFC extraction efficiency (Fig. S2b).

This study shows that the extraction capability of the hDESs vary depending on their components and can be tuned by simple mixing. In general, the extraction yields of the hDES mixtures were moderated by the individual solvents. ML and its mixtures had reasonably high extraction capability, particularly for the two classes of bioactive compounds.

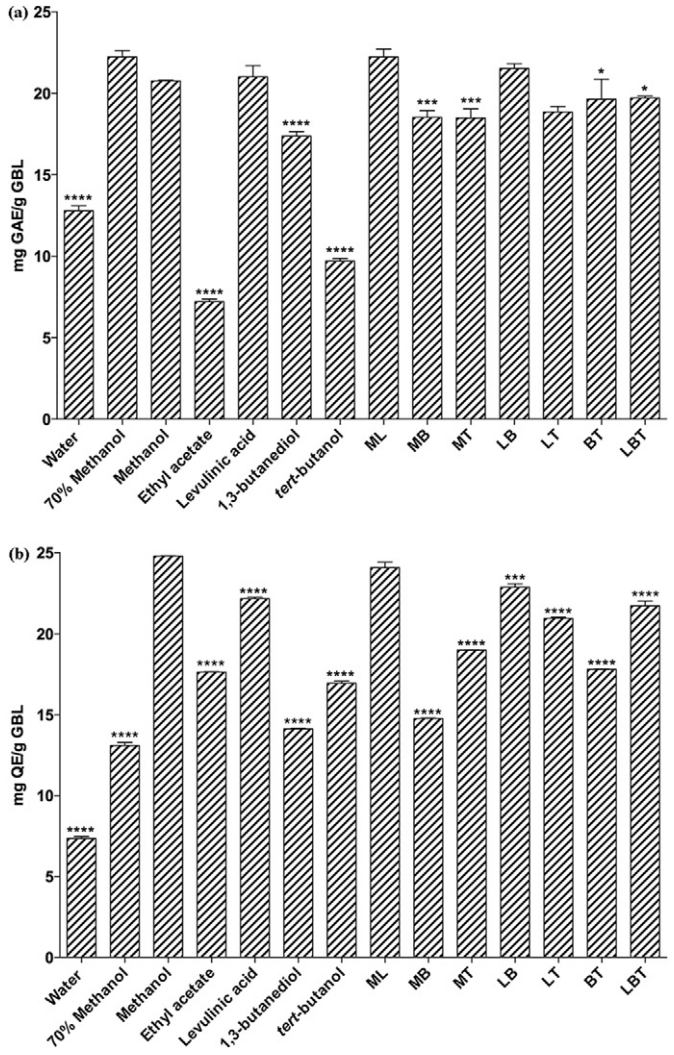


Fig. 1. Extraction capabilities of different extraction solvents in terms of TPC (a) and TFC (b). Significant differences relative to the highest value are indicated by *, **, ***, and **** for $p < 0.05$, $p < 0.01$, $p < 0.001$, and $p < 0.0001$, respectively. Error bars indicate SEM ($n = 3$). For the solvent information, see Table 1.

3.3. Extraction coverage of the menthol-based hDESs and their mixtures in non-targeted analysis

Sample preparation is important in the non-targeted analysis of a sample. The extraction solvent is the most critical factor influencing the extraction performance during sample preparation [48]. Ideally, extraction solvents should provide wide extraction coverage with good reproducibility. Herein, the GBL extracts prepared using the seven hDESs and their mixtures were applied to non-targeted analysis using UHPLC-Q-TOF/MS in both POS and NEG modes. The CV for each variable was determined from quintuplicate experiments, and only the variables with CV ≤ 30% were considered reliable with reasonable reproducibility [49,50]. In this way, the performance of the solvents for non-targeted analysis could be assessed.

Table 2 shows that all the tested solvents generally had similar extraction coverage ($NV_{CV30\%} = 160–319$) in the NEG mode. However, in the POS mode, all the hDESs and their mixtures, except for MT ($NV_{CV30\%} = 664$), presented higher extraction coverage ($NV_{CV30\%} = 902–1360$) than the reference solvents ($NV_{CV30\%} = 333–748$). The high $NV_{CV30\%}/NV_{Total}$ ratios may indicate good reproducibility of the metabolome extraction. As shown in Table 2, MB (54%) and BT (48%) exhibited the highest ratios; these values were higher than those of methanol (43%) and 70% methanol (42%) that are the extraction solvents of choice for LC-MS-based non-targeted analysis. All the other hDES-based solvents presented reasonable ratios within a narrow range (30–44%). However, the $NV_{CV30\%}/NV_{Total}$ ratio obtained with ethyl acetate was very low (20%), plausibly due to the high volatility of this solvent, leading to inconsistency in the extractant volume. Taken together, ML, MB, and their mixtures could serve as reliable extraction solvents for replacing volatile organic solvents to provide high extraction coverage with reliability for LC-MS-based non-targeted analysis.

Based on the $NV_{CV30\%}$ values, the extraction coverage of the hDES mixtures tended to be found in between those of the individual solvents, as similarly found in the polarity and extraction capability results. Then, the extraction performance of the hDES-based solvents is presented in Venn diagrams of the variables with CV ≤ 30% in Fig. 2. Although a large portion of the variables were shared between two or three hDESs, it should be noted that the extraction coverage of the hDES mixtures was not simply the average of those of the individual hDESs. For instance, LB (comprising ML and MB) extracted 242 variables that were not found in the extracts of ML or MB in the POS mode (Fig. 2). The percentage of unique variables acquired from the hDES mixtures was in the range of 6.6–22.6%. Even considering that these values are based on the variables, they are still higher than 1.3%, which is the percentage of unique metabolites extracted by the 1:1 mixture of water and methanol in comparison with pure water and pure methanol [51]. The high percentage of unique variables indicates that the extraction properties of the hDES mixtures differ from those of the constituent hDESs. Although only mixtures of equal volume ratios were tested in this study, the results suggest that mixing hDESs could yield a variety of mixtures that may act as a new kind of solvent for metabolome extraction. Thus, the properties of the hDESs for metabolome extraction could be tuned by mixing. In this manner, the limited repertoire of hDESs could be expanded very efficiently.

3.4. Extraction of GBL secondary metabolites using menthol-based hDESs and their mixtures

A large number of secondary metabolites are reportedly produced from *G. biloba*, including ginkgolides, ginkgolic acids, flavonoids, biflavones, polyphenols, and terpenoids [29]. Although some of the bioactive compounds were targeted in the hDES-based extraction [35,36], no non-targeted profiling of GBL using hDESs has been reported. In this study, the UHPLC-Q-TOF/MS-based chemical profiles of the GBL extracts were compared, and forty-seven compounds (39 from the NEG mode and 8 from the

Table 2
$NV_{CV30\%}$ values obtained by UHPLC-Q-TOF/MS-based non-targeted analysis using different extraction solvents.

Extraction solvent	NEG mode	POS mode	Total $NV_{CV30\%}$ (% in total NV)
	$NV_{CV30\%}$ (% in total NV)	$NV_{CV30\%}$ (% in total NV)	
70% methanol	244 (50%)	710 (40%)	954 (42%)
Methanol	224 (45%)	748 (42%)	972 (43%)
Ethyl acetate	160 (38%)	333 (16%)	493 (20%)
ML[a]	319 (51%)	1043 (42%)	1362 (44%)
MB [a]	180 (37%)	1360 (57%)	1540 (54%)
MT [a]	200 (43%)	664 (37%)	864 (38%)
LB [a]	228 (34%)	945 (29%)	1173 (30%)
LT [a]	305 (51%)	902 (29%)	1207 (33%)
BT [a]	283 (54%)	1035 (47%)	1318 (48%)
LBT [a]	317 (48%)	1083 (39%)	1400 (41%)

[a] For the solvent information, see Table 1.

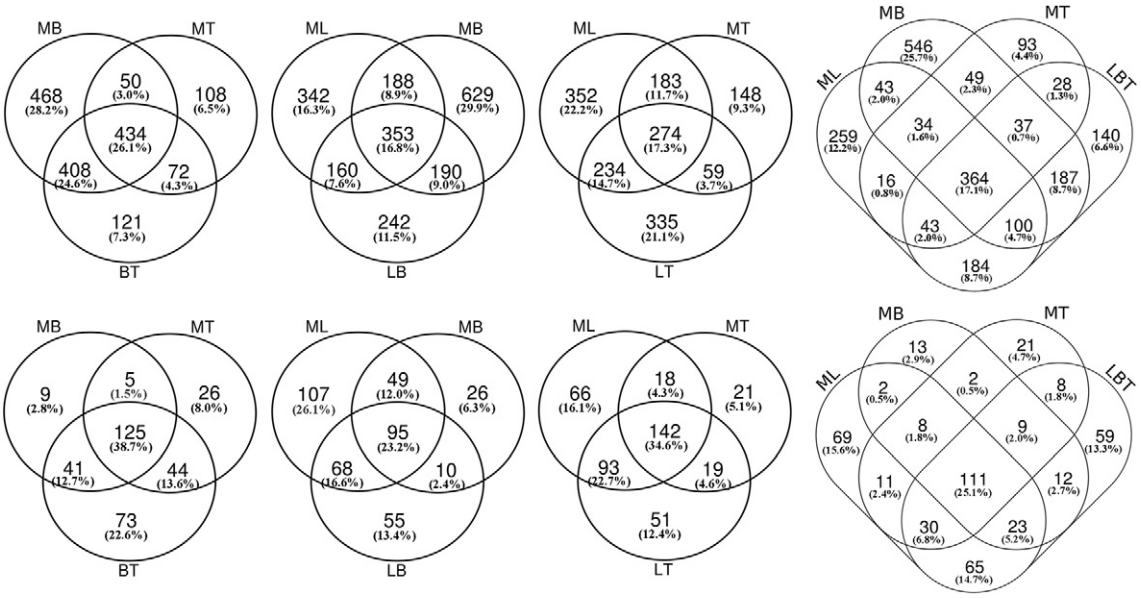


Fig. 2. Venn diagrams of the $NV_{CV30\%}$ values of different extraction solvents using non-targeted UHPLC-Q-TOF/MS analysis ($n = 5$). The top and bottom figures are from the POS and NEG modes, respectively. For the solvent information, see Table 1.

POS mode) could be identified. These compounds were mainly flavonoids; terpene trilactones, biflavones, and alkylphenols were also observed with less frequency (Table S6). Primary metabolites were not identified due to their high polarity, which made separation on the reverse stationary phase difficult. The metabolic profiles of the compounds identified after extraction with the ten solvents were subjected to HCA. The resulting dendrogram and heatmap are displayed in Fig. 3, which could help to roughly estimate the extraction specificity of the solvents within the identified compounds.

Closely clustered samples in the HCA can be regarded to have similar metabolic profiles. In Fig. 3, 70% methanol and methanol were closely grouped together and exhibited high extraction efficiency for all the metabolites, especially for flavonoids and biflavonoids. The MB and BT extracts formed a distinct cluster with the methanol-based extracts; however, they showed relatively high efficiency for flavonoids and terpene lactones, but low efficiency for biflavonoids, different from the methanol-based solvents. Given that MB had the highest extraction coverage (see Section 3.3), MB may serve as a green alternative extraction solvent for replacing methanol or 70% methanol in the LC-MS-based non-targeted analysis, with focus on the secondary metabolites.

The other five kinds of hDESs formed another major cluster, which could be further divided into two subclusters. MT and ethyl acetate, which were the most non-polar solvents in this study, formed one subcluster (Fig. 3). These solvents were relatively selective for the extraction of alkylphenols such as ginkgolic acids, which generally had long retention times (Table S6), attributed to their relatively strong hydrophobicity. Another subcluster was formed by ML and its mixtures (LT, LB, and LBT). These four solvents presented unique profiles in the heatmap. The relative intensities were generally low for all the metabolites, and the levels of alkyl phenols and terpene lactones were especially low. It is noteworthy that these ML-containing solvents showed high extraction coverage with reliability (Table 2) and high TFC and TPC extraction capability (Table S3). The apparent discrepancy might be because the extracts of the ML-containing solvents contained phenolic compounds and flavonoids that were not identified in this study. Although extraction selectivity of the solvents appeared to be associated with the solvent polarity in general, targeted analysis of various classes of compounds could provide more insights into the solvents' selectivity. Nonetheless, this study shows that the menthol-based hDESs and their mixtures with varying polarities can be used as green task-specific extraction solvents as substitutes for volatile organic solvents.

4. Conclusions

Three hDESs with different polarities, ML, MB, and MT, were selected from nine menthol-based hDESs, where the polarity was directly affected by the components. The hDESs could be mixed to produce three mixtures comprising of two hDESs and one mixture of three hDESs, the polarity of which differed from that of the individual components. The performance of the hDESs and their mixtures as extraction solvents for GBL samples was compared with that of conventional extraction solvents. ML, the most polar hDES, and its mixtures exhibited high extraction capability for phenolic compounds and flavonoids, with levels comparable to that of methanol and 70% methanol. These solvents could provide very wide extraction coverage with reliability, as reflected by their high $NV_{CV30\%}$ values and high $NV_{CV30\%}/NV_{total}$ ratios. The LC-MS-based profiles of the forty-seven secondary metabolites achieved with MB and BT were similar to those obtained with methanol and 70% methanol in the HCA and heatmap. Taken together, the present study shows that the menthol-based hDESs can be mixed to achieve properties that differ from those of the components; in this way, the repertoire of hDESs can be expanded, and therefore, the hDESs and their mixtures can be used as green, task-specific extraction solvents as substitutes for volatile organic solvents.

Author contributions

Conceptualization by Y.J. and J.L.; Investigation and Methodology by Y.J. D.J., and K.L.; Resources by D.J., K.L., and K.P.; Visualization by Y.J., D.J., K.L., and K.P.; Writing of original draft by Y.J. and J.L.; Funding acquisition and Supervision by J.L.; Writing - review & editing by J.L.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

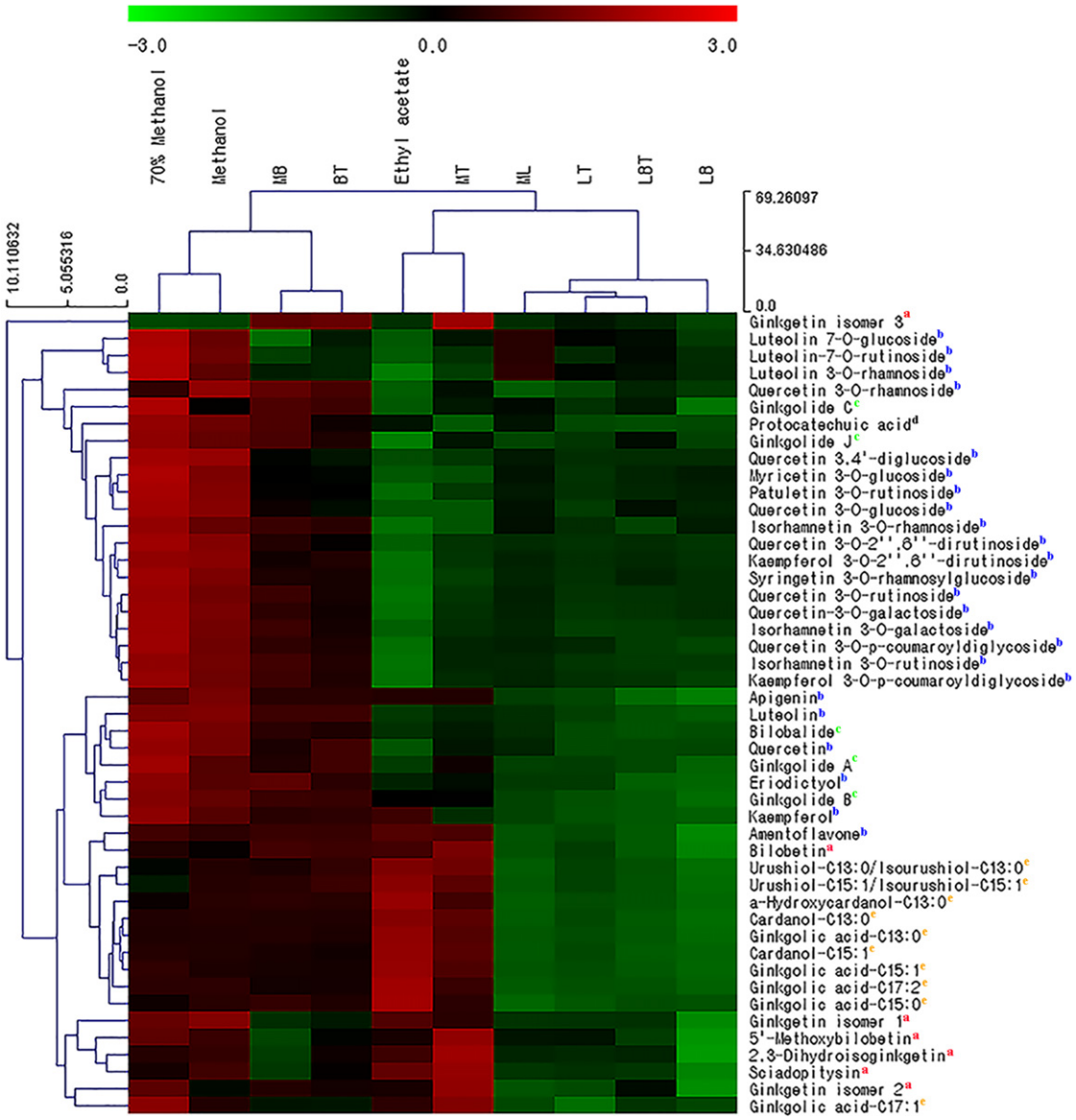


Fig. 3. Heatmap and hierarchical cluster analysis of the forty-seven identified compounds in different extraction solvents. [a] Biflavonoids; [b] flavonoids; [c] terpene lactones; [d] benzoic acids; [e] alkylphenols.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.molliq.2019.112416.

References

[1] P. Anastas, N. Eghbali, Chem. Soc. Rev. 39 (2010) 301.
[2] M.C. Bubalo, S. Vidovic, I.R. Redovnikovic, S. Jokic, J Chem Technol Biot 90 (2015) 1631.
[3] M.C. Bubalo, K. Radosevic, I.R. Redovnikovic, J. Halambek, V.G. Srcek, Ecotox Environ Safe 99 (2014) 1.
[4] J. Huang, X.Y. Guo, T.Y. Xu, L.Y. Fan, X.P. Zhou, S.H. Wu, J. Chromatogr. A 1598 (2019) 1.
[5] Y. Liu, J.B. Friesen, J.B. McAlpine, D.C. Lankin, S.N. Chen, G.F. Pauli, J. Nat. Prod. 81 (2018) 679.
[6] Y.T. Dai, G.J. Witkamp, R. Verpoorte, Y.H. Choi, Food Chem. 187 (2015) 14.
[7] M.D. Fernandez, J. Boiteux, M. Espino, F.J.V. Gomez, M.F. Silva, Anal. Chim. Acta 1038 (2018) 1.
[8] M. Ruesgas-Ramon, M.C. Figueroa-Espinoza, E. Durand, J Agr Food Chem 65 (2017) 3591.
[9] K.M. Jeong, J. Zhao, Y. Jin, S.R. Heo, S.Y. Han, D.E. Yoo, J. Lee, Arch. Pharm. Res. 38 (2015) 2143.
[10] K.M. Jeong, M. Yang, Y. Jin, E.M. Kim, J. Ko, J. Lee, Molecules 22 (2017).
[11] Y. Jin, D. Jung, K. Li, K. Park, J. Ko, M. Yang, J. Lee, Appl Sci-Basel 9 (2019).
[12] M.C. Ali, J. Chen, H.J. Zhang, Z. Li, L. Zhao, H.D. Qiu, Talanta 203 (2019) 16.
[13] K.M. Jeong, M.S. Lee, M.W. Nam, J. Zhao, Y. Jin, D.K. Lee, S.W. Kwon, J.H. Jeong, J. Lee, J. Chromatogr. A 1424 (2015) 10.
[14] B. Tang, W. Bi, H. Zhang, K.H. Row, Chromatographia 77 (2014) 373.
[15] T. Cai, H. Qiu, TrAC Trends Anal. Chem. 120 (2019), 115636.
[16] M.C. Ali, R.R. Liu, J. Chen, T.P. Cai, H.J. Zhang, Z. Li, H.L. Zhai, H.D. Qiu, Chinese Chem Lett 30 (2019) 871.
[17] H. Lee, Y. Jin, S. Kang, D. Jung, K. Park, J. Lee, Fuel In press.
[18] C. Florindo, L.C. Branco, I.M. Marrucho, Chemsuschem 12 (2019) 1549.
[19] S.M. Yousefi, F. Shemirani, S.A. Ghorbanian, J. Sep. Sci. 41 (2018) 966.
[20] A.G. Moghadam, M. Rajabi, A. Asghari, J. Chromatogr. B 1072 (2018) 50.
[21] T.G. Kazi, H.I. Afridi, M. Bhatti, A. Akhtar, Ultrason. Sonochem. 51 (2019) 40.
[22] S.Q. Zhu, J. Zhou, H.F. Jia, H.X. Zhang, Food Chem. 243 (2018) 351.
[23] J. Lee, D. Jung, K. Park, Trends Anal. Chem. 118 (2019) 853.
[24] R. Yusof, E. Abdulmalek, K. Sirat, M.B.A. Rahman, Molecules 19 (2014) 8011.
[25] D.J.G.P. van Osch, L.F. Zubeir, A. van den Bruinhorst, M.A.A. Rocha, M.C. Kroon, Green Chem. 17 (2015) 4518.
[26] J. Cao, M. Yang, F.L. Cao, J.H. Wang, E.Z. Su, J. Clean. Prod. 152 (2017) 399.

[27] M.B. Haider, D. Jha, R. Kumar, B.M. Sivagnanam, International Journal of Greenhouse Gas Control In press (2019).
[28] J. Cao, M. Yang, F.L. Cao, J.H. Wang, E.Z. Su, ACS Sustain. Chem. Eng. 5 (2017) 3270.
[29] T.A. van Beek, P. Montoro, J. Chromatogr. A 1216 (2009) 2002.
[30] N. Lopez-Gutierrez, R. Rornero-Gonzalez, J.L.M. Vidal, A.G. Frenich, J Pharmaceut Biomed 121 (2016) 151.
[31] J. Cao, H.M. Wang, W. Zhang, F.L. Cao, G.L. Ma, E.Z. Su, Molecules 23 (2018).
[32] M. Yang, J. Cao, F. Cao, C. Lu, E. Su, Chem. Biochem. Eng. Q. 32 (2018) 315.
[33] J. Cao, L.Y. Chen, M.H. Li, F.L. Cao, L.G. Zhao, E.Z. Su, J Pharmaceut Biomed 158 (2018) 317.
[34] E.Z. Su, M. Yang, J. Cao, C. Lu, J.H. Wang, F.L. Cao, J Liq Chromatogr R T 40 (2017) 385.
[35] J. Cao, L.Y. Chen, M.H. Li, F.L. Cao, L.G. Zhao, E.Z. Su, Green Chem. 20 (2018) 1879.
[36] L.-T. Wang, Q. Yang, Q. Cui, X.-H. Fan, M.-Z. Dong, M.-Z. Gao, M.-J. Lv, J.-Y. An, D. Meng, X.-H. Zhao, Y.-J. Fu, J. Clean. Prod. In press (2019).
[37] D.E. Yoo, K.M. Jeong, S.Y. Han, E.M. Kim, Y. Jin, J. Lee, Food Chem. 255 (2018) 357.
[38] Y.T. Dai, J. van Spronsen, G.J. Witkamp, R. Verpoorte, Y.H. Choi, Anal. Chim. Acta 766 (2013) 61.
[39] K.M. Jeong, J. Ko, J. Zhao, Y. Jin, D.E. Yoo, S.Y. Han, J. Lee, J. Clean. Prod. 151 (2017) 87.
[40] Komal, G. Singh, G. Singh, T.S. Kang, Acs Omega 3 (2018) 13387.
[41] K.M. Jeong, Y. Jin, D.E. Yoo, S.Y. Han, E.M. Kim, J. Lee, Food Chem. 251 (2018) 69.
[42] L. Chupin, C. Motillon, F. Charrier-El Bouhtoury, A. Pizzi, B. Charrier, Ind. Crop. Prod. 49 (2013) 897.
[43] Y. Jin, J. Zhao, E.M. Kim, K.H. Kim, S. Kang, H. Lee, J. Lee, Molecules 24 (2019).
[44] B.D. Ribeiro, C. Florindo, L.C. Iff, M.A.Z. Coelho, I.M. Marrucho, ACS Sustain. Chem. Eng. 3 (2015) 2469.
[45] H.N. Ngoc, D.T. Nghiem, T.L.G. Pham, H. Stuppner, M. Ganzera, J Pharmaceut Biomed 149 (2018) 66.
[46] D.Y. Gu, Y. Yang, M. Bakri, Q.B. Chen, H.A. Aisa, Nat. Prod. Res. 32 (2018) 2054.
[47] H.D.F.Q. Barros, A.M. Baseggio, C.F.F. Angolini, G.M. Pastore, C.B.B. Cazarin, M.R. Marostica-Junior, Food Res. Int. 124 (2019) 16.
[48] O. Deda, A.C. Chatziioannou, S. Fasoula, D. Palachanis, N. Raikos, G.A. Theodoridis, H.G. Gika, J. Chromatogr. B 1047 (2017) 115.
[49] M.A. Anwar, P.A. Vorkas, J.V. Li, J. Shalhoub, E.J. Want, A.H. Davies, E. Holmes, Analyst 140 (2015) 7586.
[50] K. Spagou, I.D. Wilson, P. Masson, G. Theodoridis, N. Raikos, M. Coen, E. Holmes, J.C. Lindon, R.S. Plumb, J.K. Nicholson, E.J. Want, Anal. Chem. 83 (2011) 382.
[51] M.A. Lopez-Bascon, M. Calderon-Santiago, H. Arguello, L. Morera, J.J. Garrido, F. Priego-Capote, Talanta 199 (2019) 303.